Deploying a Scalable Web Application Using AWS Elastic Beanstalk

📘 Scenario Description:

A startup company aims to simplify application deployment and management while ensuring scalability and reliability. To achieve this, the team will leverage AWS Elastic Beanstalk, which abstracts the complexity of infrastructure provisioning, application deployment, and auto scaling.

Scenario Steps:

Setting Up the Beanstalk Environment:

1- Create a Web Application Package
Develop a simple web application using Node.js, Python (Flask), or PHP.
Package the code and dependencies in a .zip file as per AWS Beanstalk requirements.

2- Deploy the Application using Elastic Beanstalk Console
Create a new Elastic Beanstalk environment (Web server environment).
Choose a platform (e.g., Node.js, Python, PHP).
Upload the application package and deploy.

3- Enable Auto Scaling
Use the Elastic Beanstalk configuration options to enable automatic scaling based on CPU or request count.
Configure minimum and maximum instance counts.

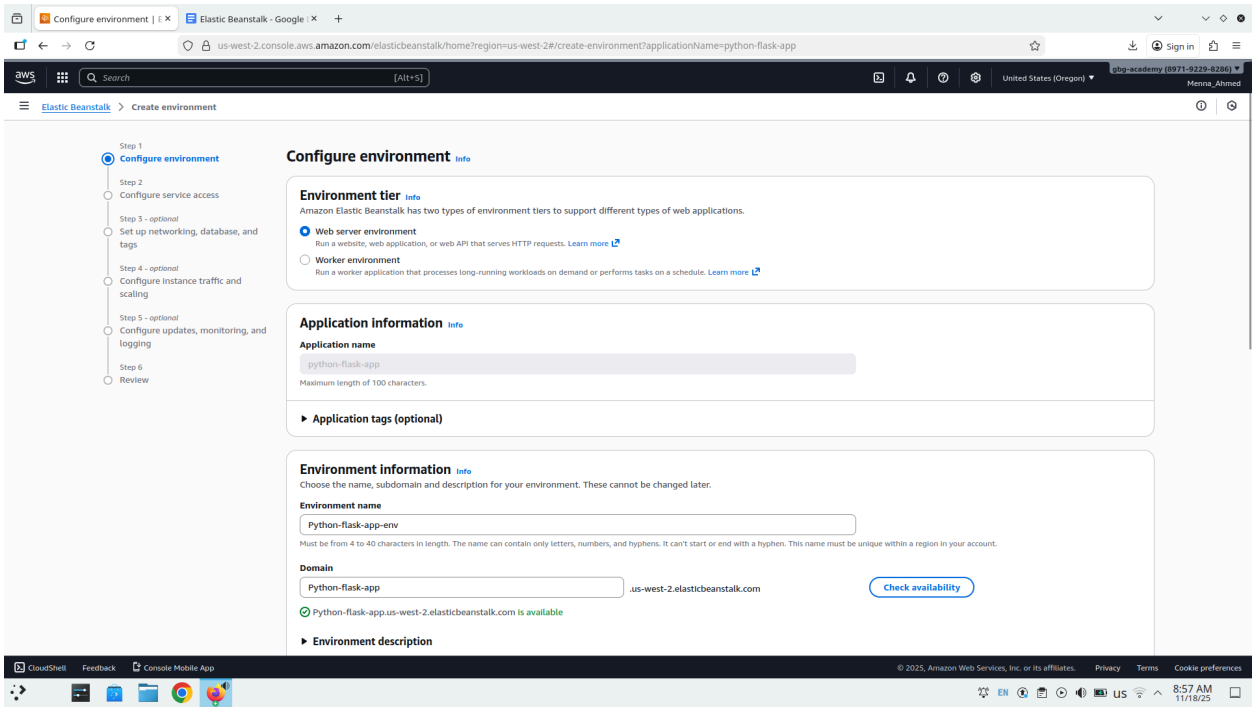
4- Attach Load Balancer
Elastic Beanstalk automatically creates and configures an Application Load Balancer (ALB).
Validate health checks and traffic routing.

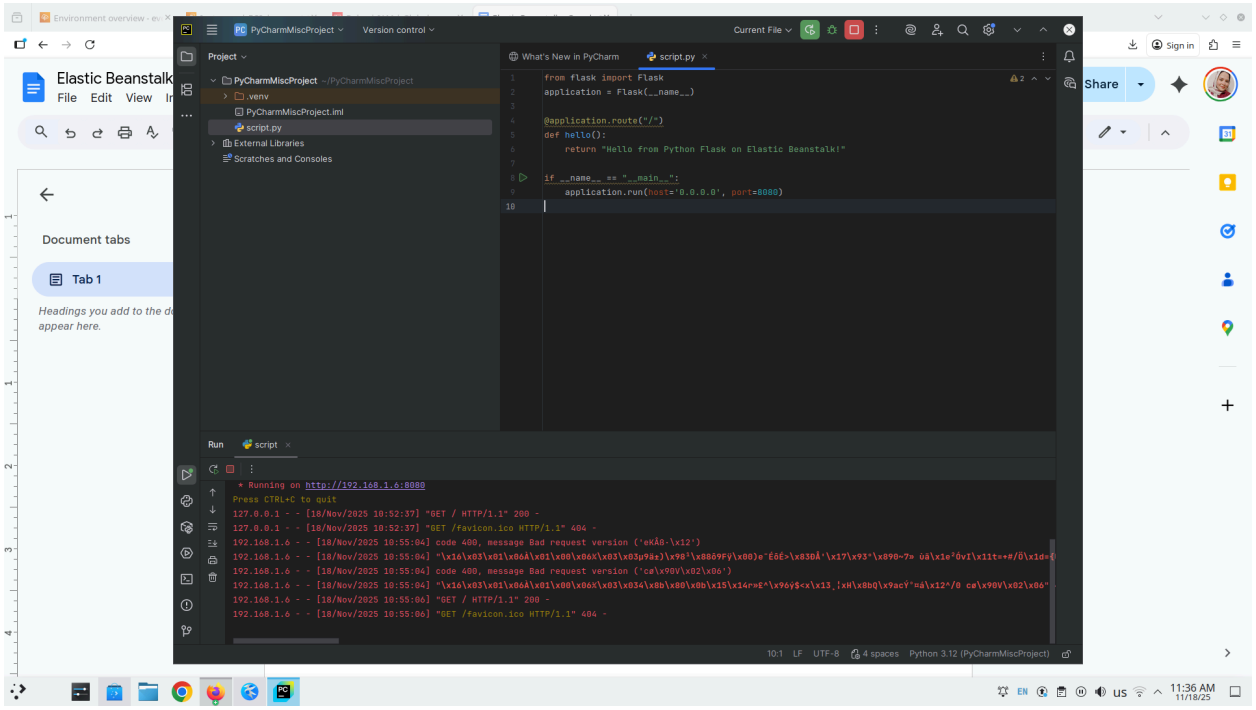
Scenario Testing:

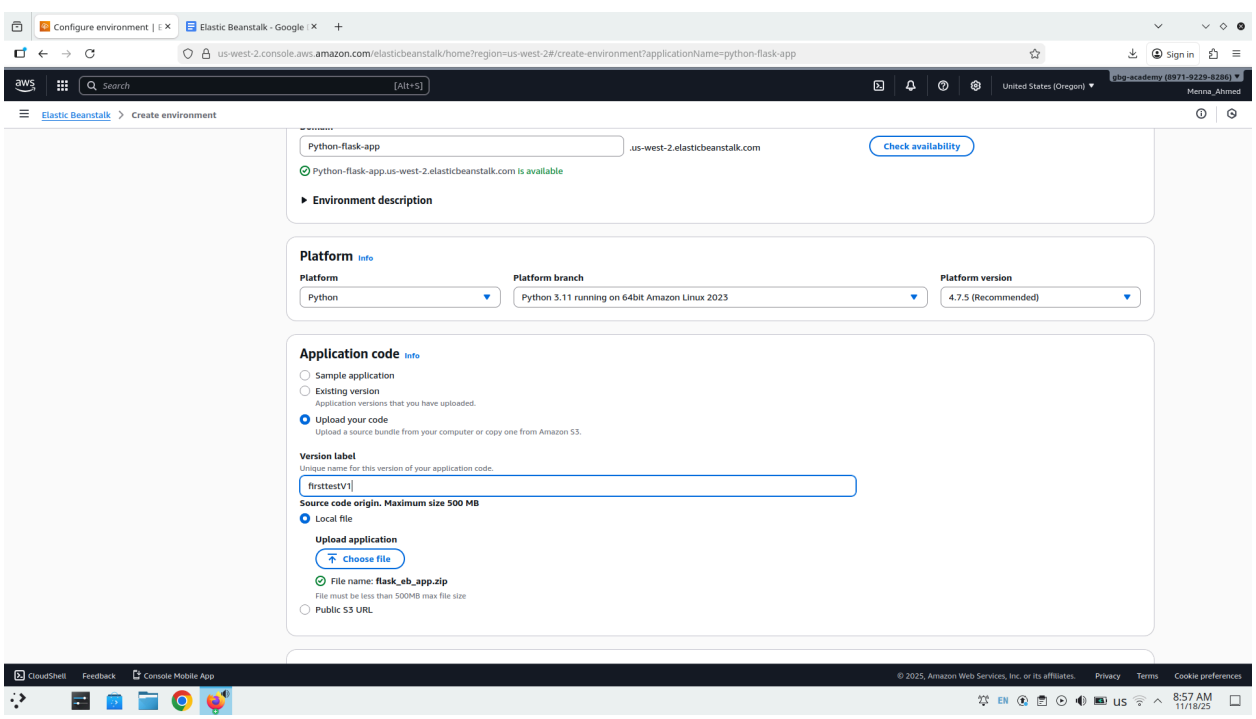
Run Application Functionality Tests

Go to console and let’s start:-

create Elastic BeanStalk




PyCharmMiscProject
Version control
Current File
Project
PyCharmMiscProject ~/PyCharmMiscProject
.venv
PyCharmMiscProject.iml
script.py
External Libraries
Scratches and Consoles
What's New in PyCharm
script.py
from flask import Flask
application = Flask(__name__)
@application.route("/")
def hello():
return "Hello from Python Flask on Elastic Beanstalk!"
if __name__ == "__main__":
application.run(host='0.0.0.0', port=8080)
Run
script
* Running on http://192.168.1.6:8080
Press CTRL+C to quit
127.0.0.1 - - [18/Nov/2025 10:52:37] "GET / HTTP/1.1" 200 -
127.0.0.1 - - [18/Nov/2025 10:52:37] "GET /favicon.ico HTTP/1.1" 404 -
192.168.1.6 - - [18/Nov/2025 10:55:04] code 400, message Bad request version ('eKÃ0-\x12')
192.168.1.6 - - [18/Nov/2025 10:55:04] code 400, message Bad request version ('ce\x90V\x02\x06')
192.168.1.6 - - [18/Nov/2025 10:55:06] "GET / HTTP/1.1" 200 -
192.168.1.6 - - [18/Nov/2025 10:55:06] "GET /favicon.ico HTTP/1.1" 404 -
10:1 LF UTF-8 4 spaces Python 3.12 (PyCharmMiscProject)
Elastic Beanstalk
File Edit View
Share
Document tabs
Tab 1
Headings you add to the do
appear here.
11:36 AM
11/18/25


Configure environment
Elastic Beanstalk - Google
us-west-2.console.aws.amazon.com/elasticbeanstalk/home?region=us-west-2#/create-environment?applicationName=python-flask-app
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Elastic Beanstalk > Create environment
Python-flask-app
.us-west-2.elasticbeanstalk.com
Check availability
Python-flask-app.us-west-2.elasticbeanstalk.com is available
Environment description
Platform Info
Platform
Python
Platform branch
Python 3.11 running on 64bit Amazon Linux 2023
Platform version
4.7.5 (Recommended)
Application code Info
Sample application
Existing version
Application versions that you have uploaded.
Upload your code
Upload a source bundle from your computer or copy one from Amazon S3.
Version label
Unique name for this version of your application code.
firsttestV1
Source code origin. Maximum size 500 MB
Local file
Upload application
Choose file
File name: flask_eb_app.zip
File must be less than 500MB max file size
Public S3 URL
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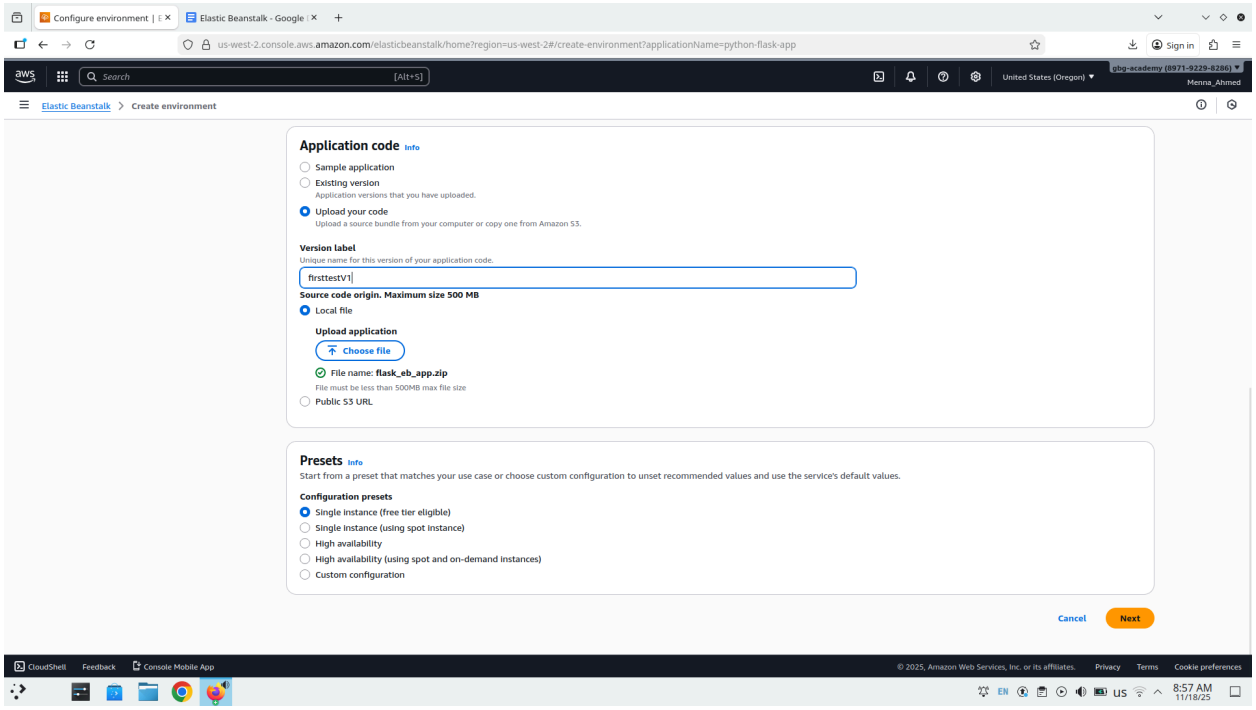
Configure environment
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us-west-2.console.aws.amazon.com/elasticbeanstalk/home?region=us-west-2#/create-environment?applicationName=python-flask-app
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Elastic Beanstalk
Create environment
Application code Info
Sample application
Existing version
Application versions that you have uploaded.
Upload your code
Upload a source bundle from your computer or copy one from Amazon S3.
Version label
Unique name for this version of your application code.
firsttestV1
Source code origin. Maximum size 500 MB
Local file
Upload application
Choose file
File name: flask_eb_app.zip
File must be less than 500MB max file size
Public S3 URL
Presets Info
Start from a preset that matches your use case or choose custom configuration to unset recommended values and use the service's default values.
Configuration presets
Single instance (free tier eligible)
Single instance (using spot instance)
High availability
High availability (using spot and on-demand instances)
Custom configuration
Cancel
Next
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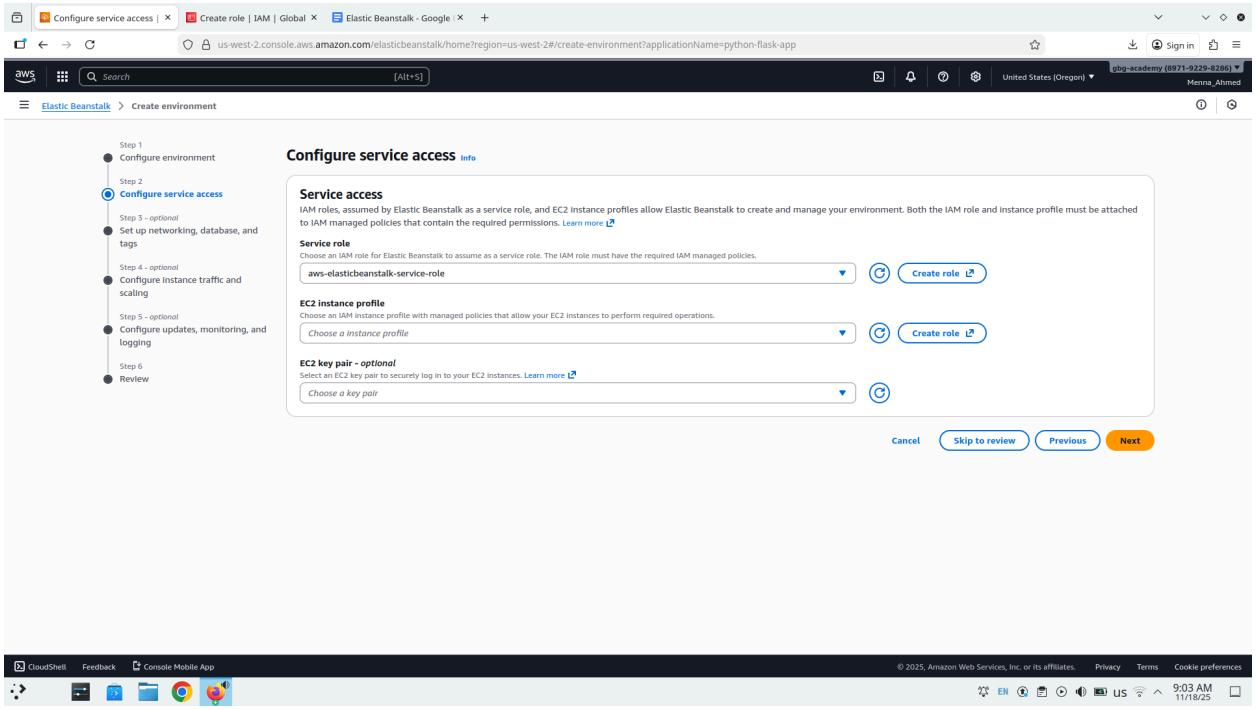
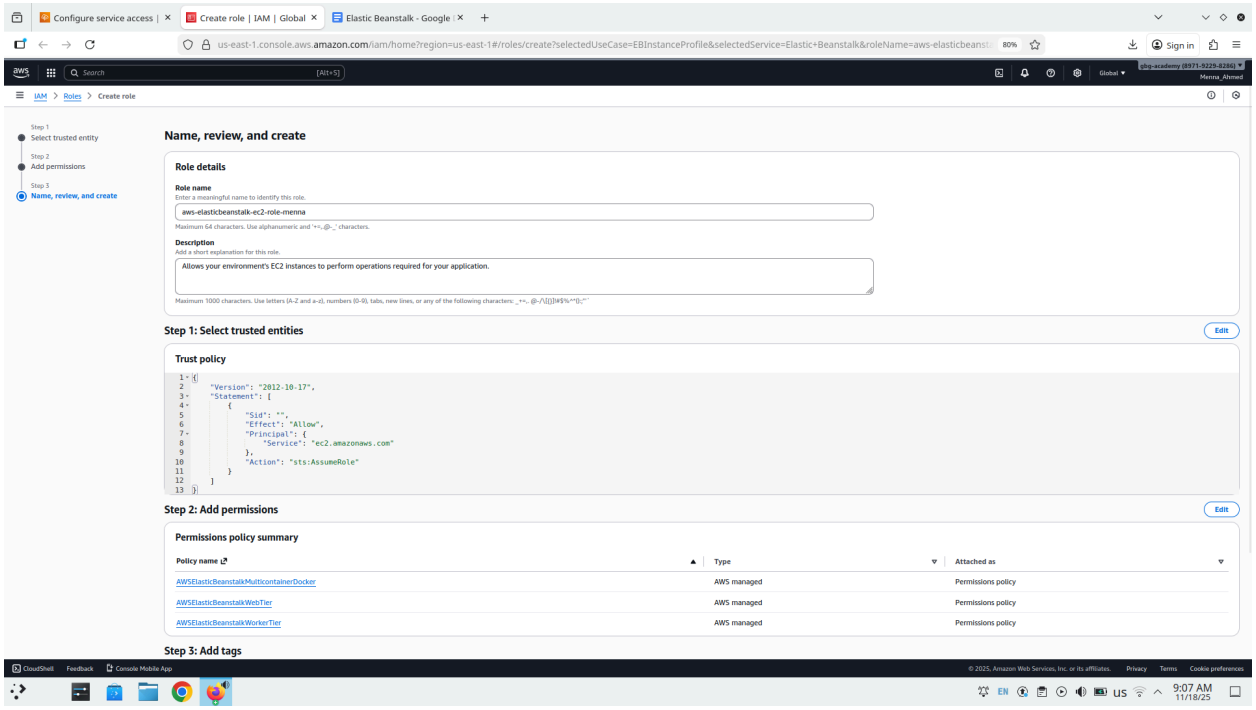
Next



I will create IAM Role and give role to Elastic beanstalk



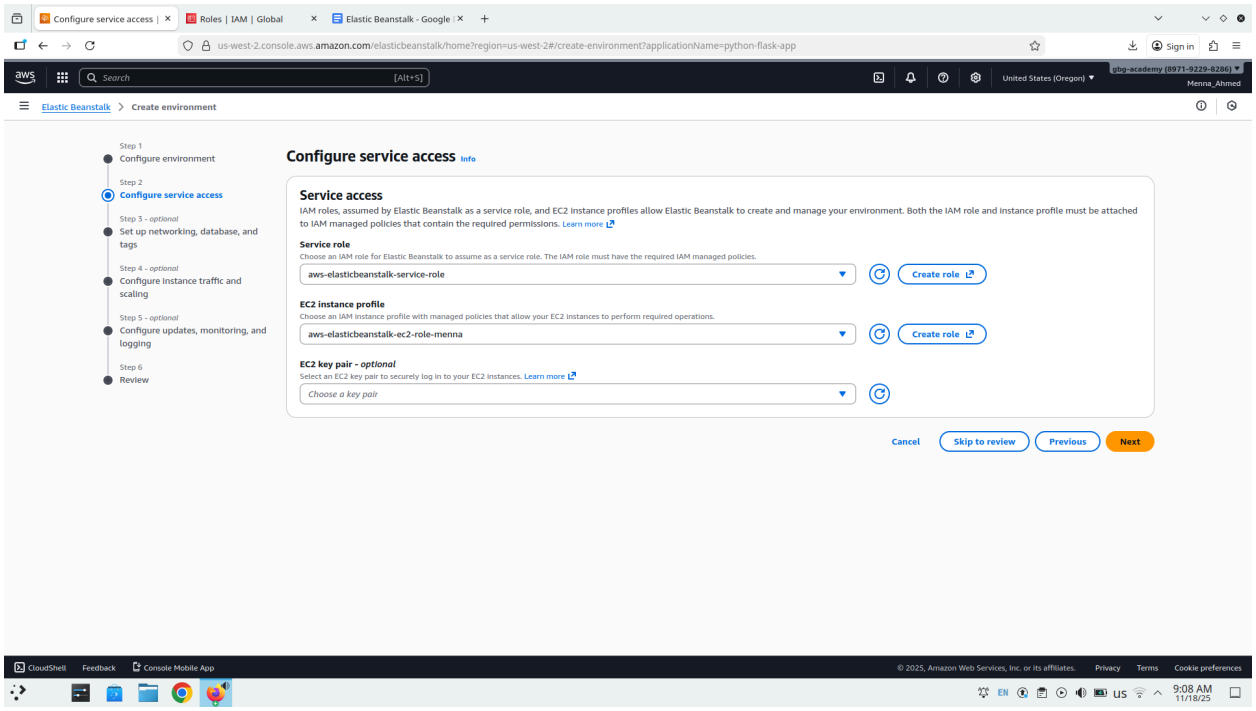
3 AWS managed policies are the required default policies for the EC2 instance in Elastic Beanstalk:

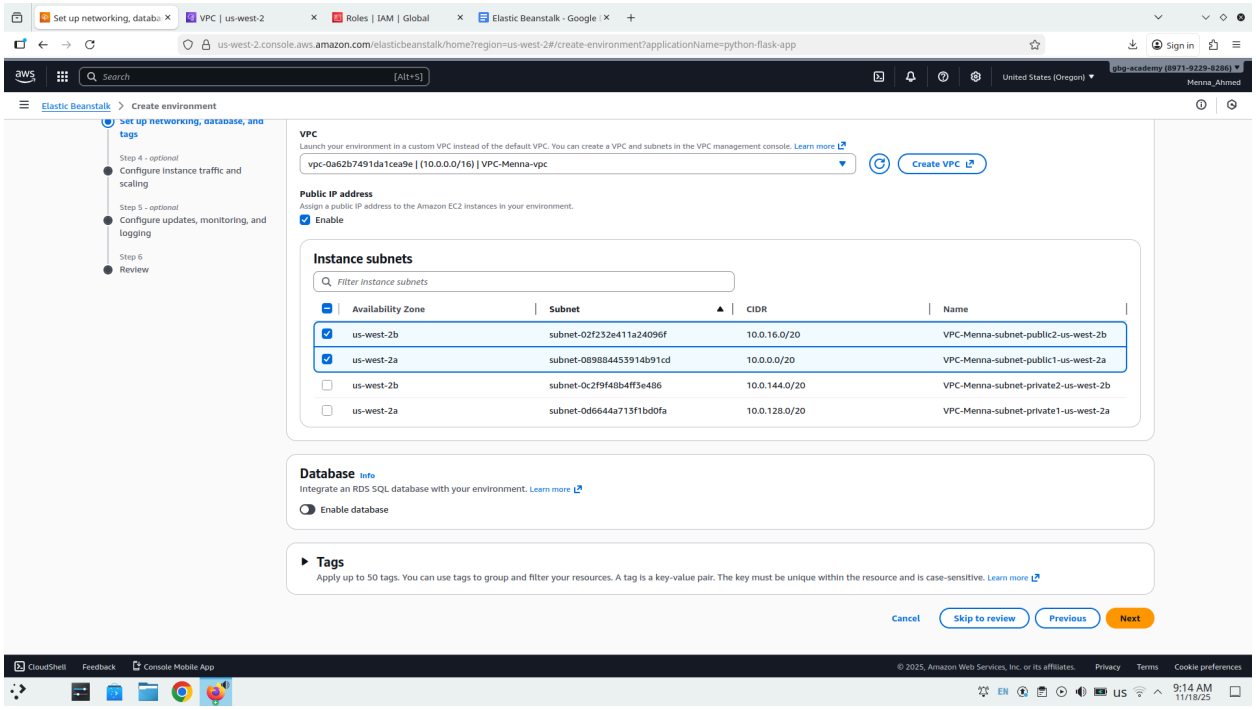
- AWSElasticBeanstalkMulticontainerDocker
- AWSElasticBeanstalkWebTier
- AWSElasticBeanstalkWorkerTier

These policies give the EC2 instance the right permissions to:

- Download your application from S3
- Write logs to CloudWatch
- Communicate status to Elastic Beanstalk
- Run your web app container
- Handle updates and health checks

Back to continue processing





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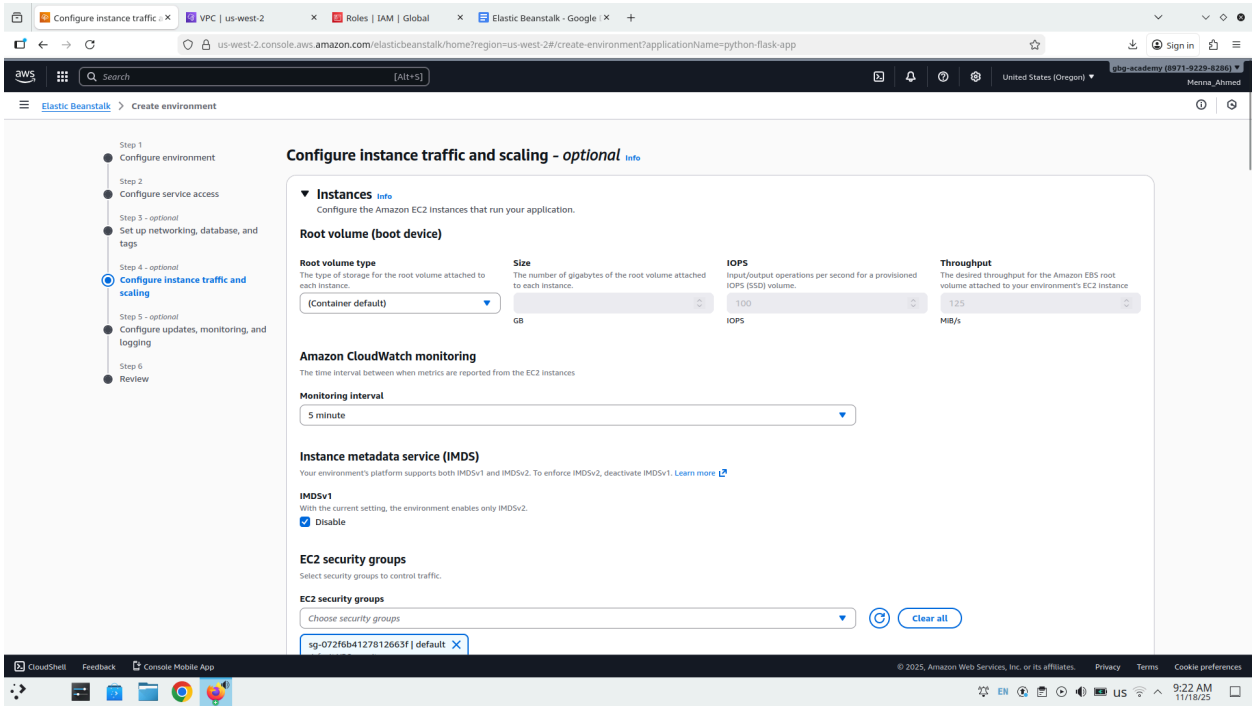
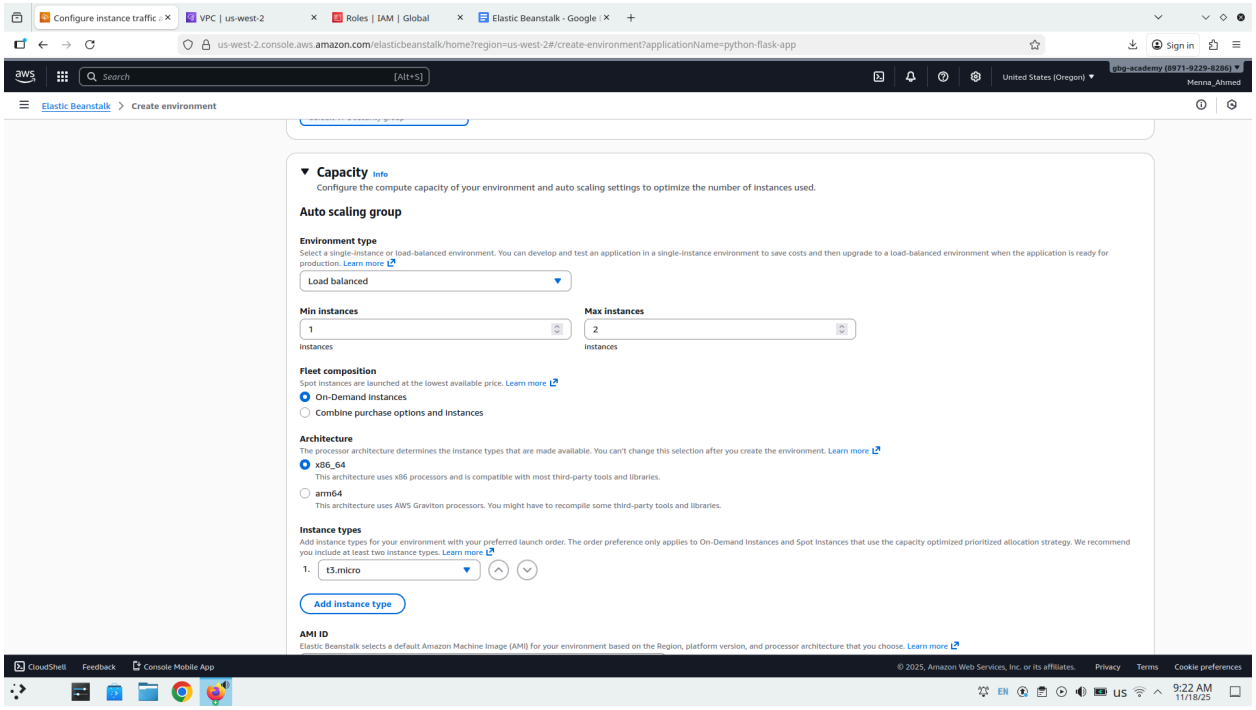
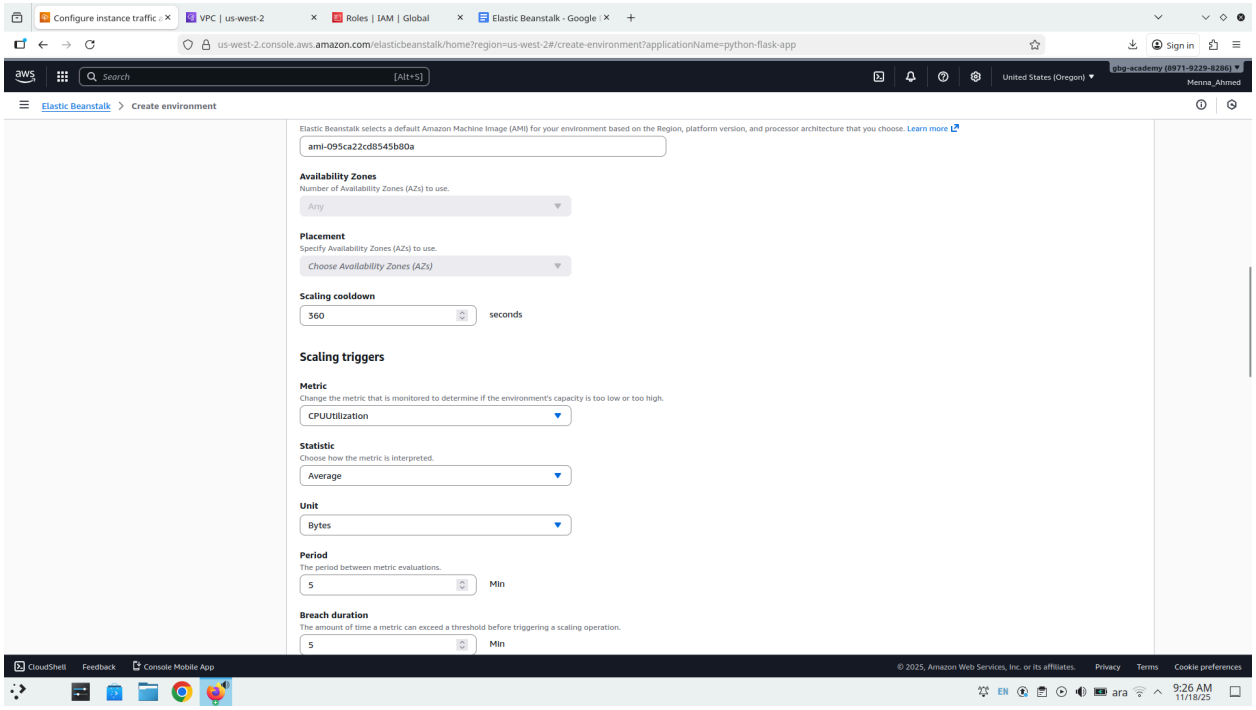

Configure instance traffic
VPC | us-west-2
Roles | IAM | Global
Elastic Beanstalk - Google
us-west-2.console.aws.amazon.com/elasticbeanstalk/home?region=us-west-2#/create-environment?applicationName=python-flask-app
Elastic Beanstalk > Create environment
Step 1 Configure environment
Step 2 Configure service access
Step 3 - optional Set up networking, database, and tags
Step 4 - optional Configure instance traffic and scaling
Step 5 - optional Configure updates, monitoring, and logging
Step 6 Review
Configure instance traffic and scaling - optional Info
Instances Info
Configure the Amazon EC2 instances that run your application.
Root volume (boot device)
Root volume type
(Container default)
Size
IOPS
Throughput
Amazon CloudWatch monitoring
Monitoring interval
5 minute
Instance metadata service (IMDS)
IMDSv1
Disable
EC2 security groups
Choose security groups
Clear all
sg-072f6b4127812663f | default


Elastic Beanstalk > Create environment
Capacity Info
Configure the compute capacity of your environment and auto scaling settings to optimize the number of instances used.
Auto scaling group
Environment type
Load balanced
Min instances
1
Max instances
2
Fleet composition
On-Demand instances
Combine purchase options and instances
Architecture
x86_64
arm64
Instance types
t3.micro
Add instance type
AMI ID


Configure instance traffic
VPC | us-west-2
Roles | IAM | Global
Elastic Beanstalk - Google
us-west-2.console.aws.amazon.com/elasticbeanstalk/home?region=us-west-2#/create-environment?applicationName=python-flask-app
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Elastic Beanstalk > Create environment
Elastic Beanstalk selects a default Amazon Machine Image (AMI) for your environment based on the Region, platform version, and processor architecture that you choose. Learn more
ami-095ca22cd8545b80a
Availability Zones
Number of Availability Zones (AZs) to use.
Any
Placement
Specify Availability Zones (AZs) to use.
Choose Availability Zones (AZs)
Scaling cooldown
360
seconds
Scaling triggers
Metric
Change the metric that is monitored to determine if the environment's capacity is too low or too high.
CPUUtilization
Statistic
Choose how the metric is interpreted.
Average
Unit
Bytes
Period
The period between metric evaluations.
5
Min
Breach duration
The amount of time a metric can exceed a threshold before triggering a scaling operation.
5
Min
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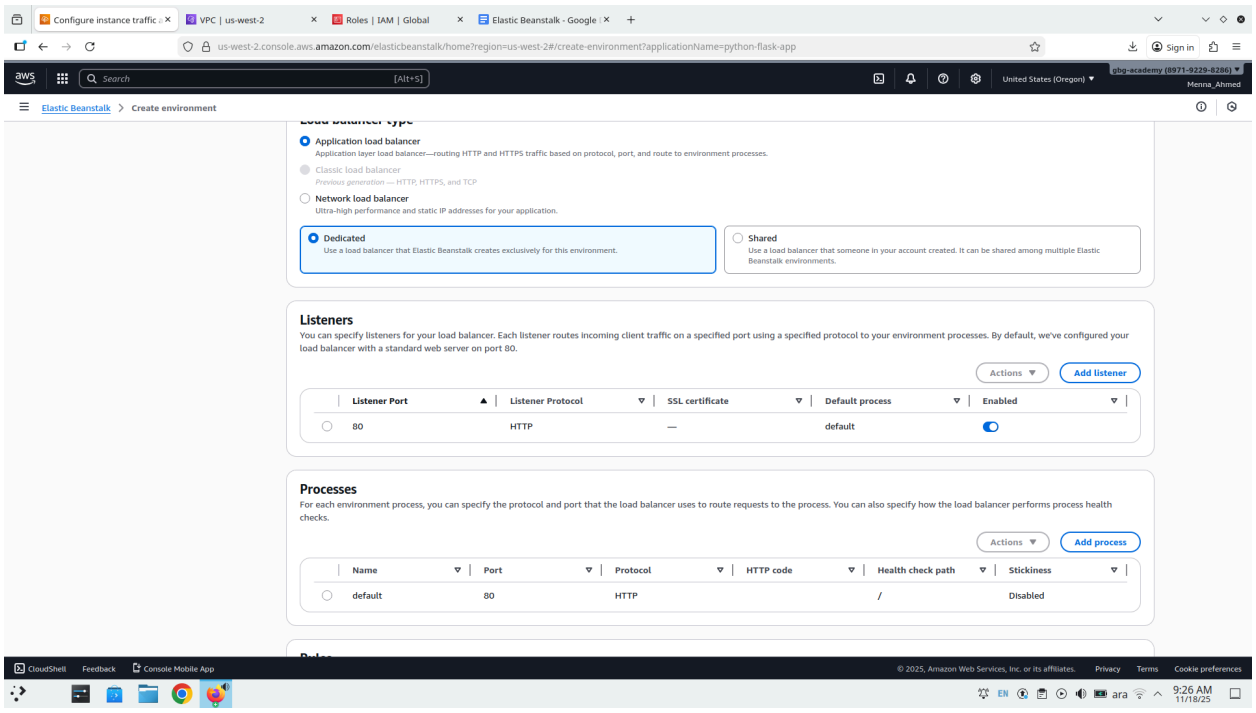
Configure instance traffic
VPC | us-west-2
Roles | IAM | Global
Elastic Beanstalk - Google
us-west-2.console.aws.amazon.com/elasticbeanstalk/home?region=us-west-2#/create-environment?applicationName=python-flask-app
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Elastic Beanstalk > Create environment
Application load balancer
Application layer load balancer—routing HTTP and HTTPS traffic based on protocol, port, and route to environment processes.
Classic load balancer
Previous generation — HTTP, HTTPS, and TCP
Network load balancer
Ultra-high performance and static IP addresses for your application.
Dedicated
Use a load balancer that Elastic Beanstalk creates exclusively for this environment.
Shared
Use a load balancer that someone in your account created. It can be shared among multiple Elastic Beanstalk environments.
Listeners
You can specify listeners for your load balancer. Each listener routes incoming client traffic on a specified port using a specified protocol to your environment processes. By default, we've configured your load balancer with a standard web server on port 80.
Actions
Add listener
Listener Port
Listener Protocol
SSL certificate
Default process
Enabled
80
HTTP
—
default
Processes
For each environment process, you can specify the protocol and port that the load balancer uses to route requests to the process. You can also specify how the load balancer performs process health checks.
Actions
Add process
Name
Port
Protocol
HTTP code
Health check path
Stickiness
default
80
HTTP
/
Disabled
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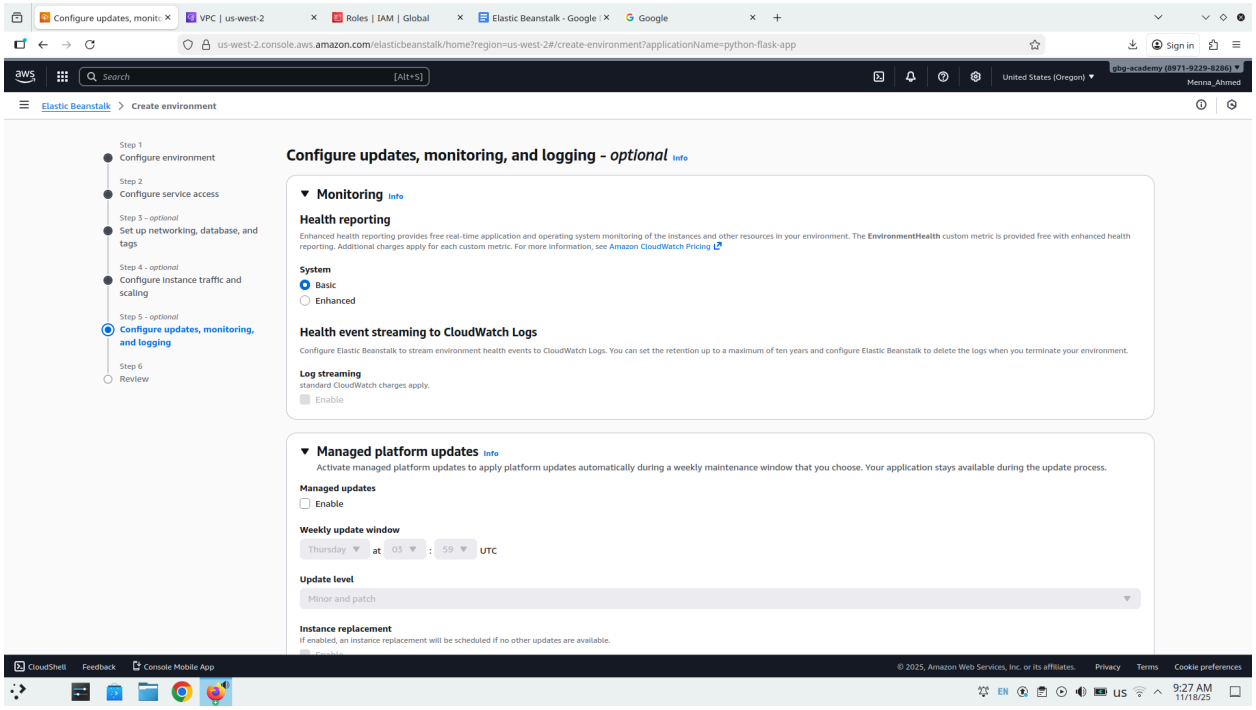
Configure updates, monitor
VPC | us-west-2
Roles | IAM | Global
Elastic Beanstalk - Google
Google
us-west-2.console.aws.amazon.com/elasticbeanstalk/home?region=us-west-2#/create-environment?applicationName=python-flask-app
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Elastic Beanstalk > Create environment
Step 1
Configure environment
Step 2
Configure service access
Step 3 - optional
Set up networking, database, and tags
Step 4 - optional
Configure instance traffic and scaling
Step 5 - optional
Configure updates, monitoring, and logging
Step 6
Review
Configure updates, monitoring, and logging - optional Info
Monitoring Info
Health reporting
Enhanced health reporting provides free real-time application and operating system monitoring of the instances and other resources in your environment. The EnvironmentHealth custom metric is provided free with enhanced health reporting. Additional charges apply for each custom metric. For more information, see Amazon CloudWatch Pricing
System
Basic
Enhanced
Health event streaming to CloudWatch Logs
Configure Elastic Beanstalk to stream environment health events to CloudWatch Logs. You can set the retention up to a maximum of ten years and configure Elastic Beanstalk to delete the logs when you terminate your environment.
Log streaming
standard CloudWatch charges apply.
Enable
Managed platform updates Info
Activate managed platform updates to apply platform updates automatically during a weekly maintenance window that you choose. Your application stays available during the update process.
Managed updates
Enable
Weekly update window
Thursday
at
03
59
UTC
Update level
Minor and patch
Instance replacement
If enabled, an instance replacement will be scheduled if no other updates are available.
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Console Mobile App
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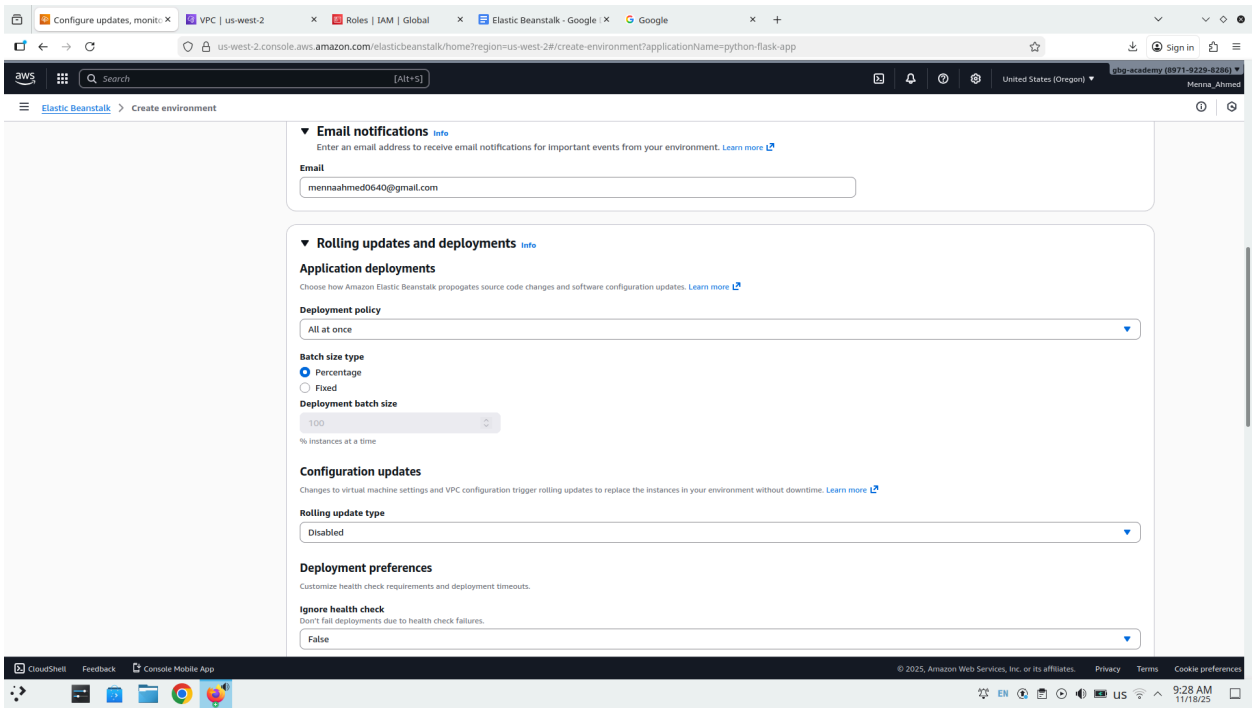
Configure updates, monitor
VPC | us-west-2
Roles | IAM | Global
Elastic Beanstalk - Google
Google
us-west-2.console.aws.amazon.com/elasticbeanstalk/home?region=us-west-2#/create-environment?applicationName=python-flask-app
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Elastic Beanstalk > Create environment
Email notifications Info
Enter an email address to receive email notifications for important events from your environment. Learn more
Email
mennaahmed0640@gmail.com
Rolling updates and deployments Info
Application deployments
Choose how Amazon Elastic Beanstalk propogates source code changes and software configuration updates. Learn more
Deployment policy
All at once
Batch size type
Percentage
Fixed
Deployment batch size
100
% instances at a time
Configuration updates
Changes to virtual machine settings and VPC configuration trigger rolling updates to replace the instances in your environment without downtime. Learn more
Rolling update type
Disabled
Deployment preferences
Customize health check requirements and deployment timeouts.
Ignore health check
Don't fail deployments due to health check failures.
False
CloudShell
Feedback
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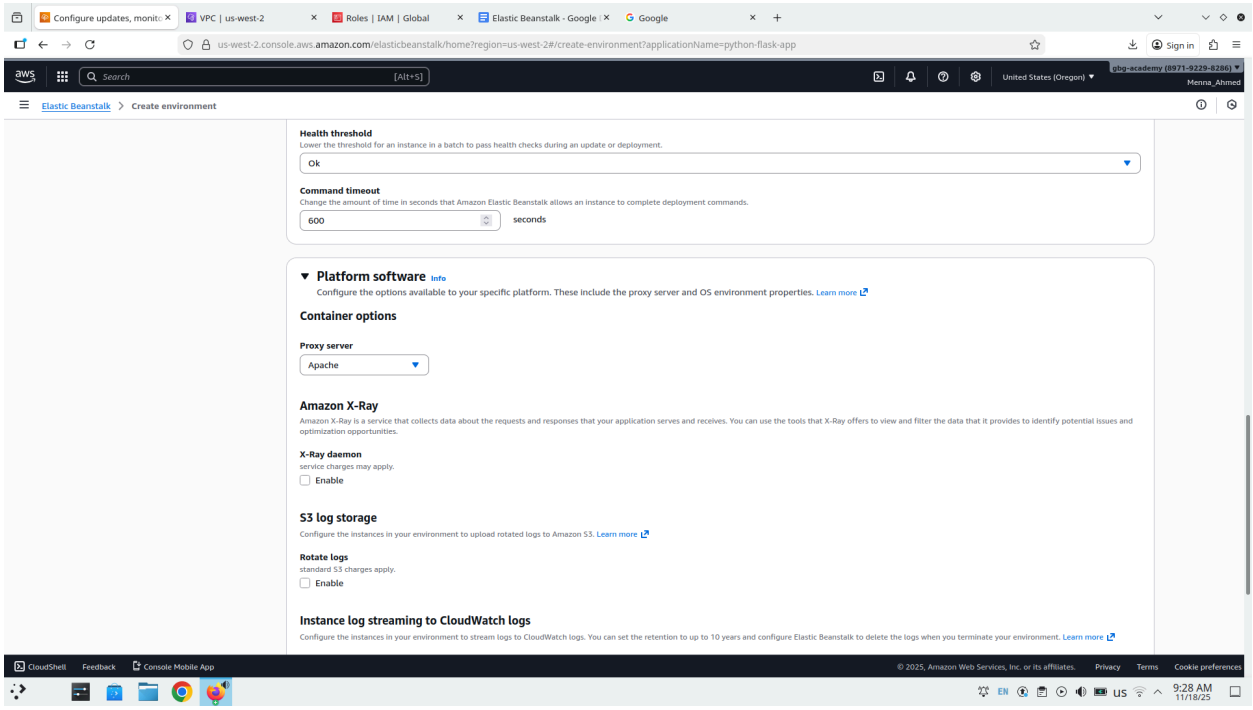
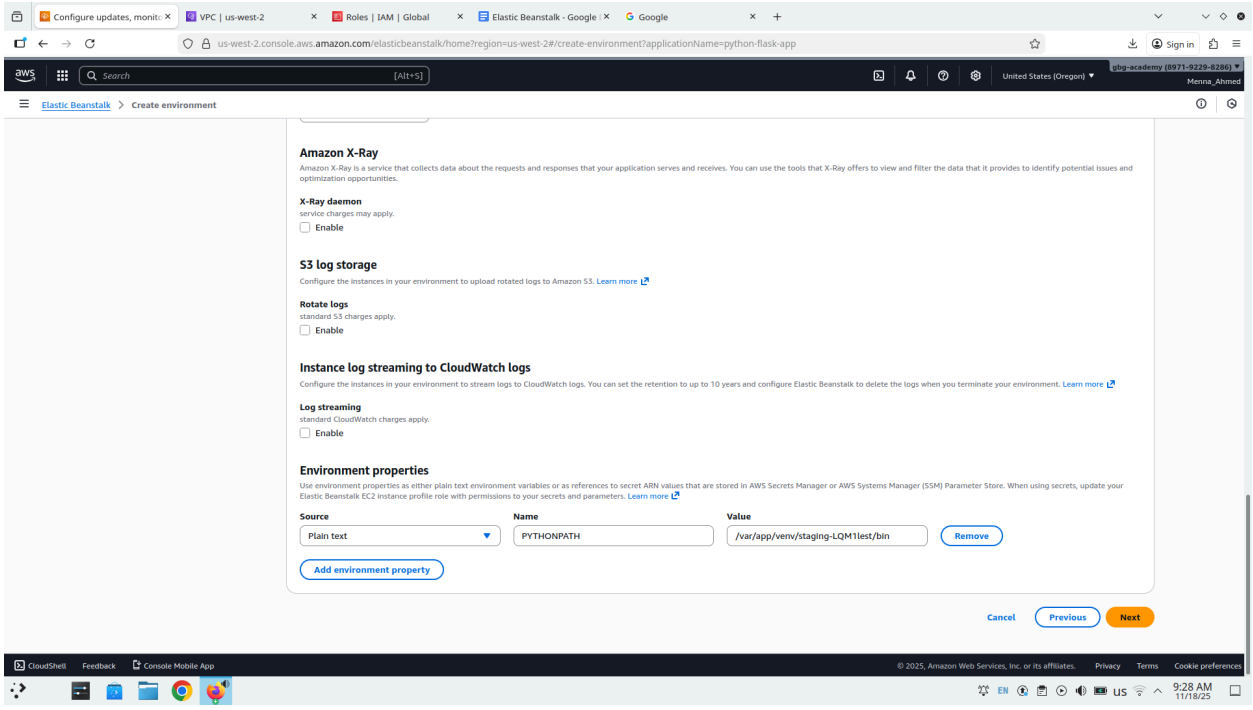

Elastic Beanstalk > Create environment
Health threshold
Lower the threshold for an instance in a batch to pass health checks during an update or deployment.
Ok
Command timeout
Change the amount of time in seconds that Amazon Elastic Beanstalk allows an instance to complete deployment commands.
600
seconds
Platform software Info
Configure the options available to your specific platform. These include the proxy server and OS environment properties. Learn more
Container options
Proxy server
Apache
Amazon X-Ray
Amazon X-Ray is a service that collects data about the requests and responses that your application serves and receives. You can use the tools that X-Ray offers to view and filter the data that it provides to identify potential issues and optimization opportunities.
X-Ray daemon
service charges may apply.
Enable
S3 log storage
Configure the instances in your environment to upload rotated logs to Amazon S3. Learn more
Rotate logs
standard S3 charges apply.
Enable
Instance log streaming to CloudWatch logs
Configure the instances in your environment to stream logs to CloudWatch logs. You can set the retention to up to 10 years and configure Elastic Beanstalk to delete the logs when you terminate your environment. Learn more


Elastic Beanstalk > Create environment
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Instance log streaming to CloudWatch logs
Configure the instances in your environment to stream logs to CloudWatch logs. You can set the retention to up to 10 years and configure Elastic Beanstalk to delete the logs when you terminate your environment. Learn more
Log streaming
standard CloudWatch charges apply.
Enable
Environment properties
Use environment properties as either plain text environment variables or as references to secret ARN values that are stored in AWS Secrets Manager or AWS Systems Manager (SSM) Parameter Store. When using secrets, update your Elastic Beanstalk EC2 instance profile role with permissions to your secrets and parameters. Learn more
Source
Plain text
Name
PYTHONPATH
Value
/var/app/venv/staging-LQM1est/bin
Remove
Add environment property
Cancel
Previous
Next

Then next

Overview

Elastic Beanstalk > Create environment

Step 1 Configure environment
Step 2 Configure service access
Step 3 - optional Set up networking, database, and tags
Step 4 - optional Configure instance traffic and scaling
Step 5 - optional Configure updates, monitoring, and logging
Step 6 Review

Review Info

Step 1: Configure environment

Edit

Environment information

Environment tier Web server environment	**Application name** python-flask-app
Environment name Python-flask-app-env	**Application code** flask_eb_app.zip

Platform
arn:aws:elasticbeanstalk:us-west-2::platform/Python 3.11 running on 64bit Amazon Linux 2023/4.7.5

Step 2: Configure service access

Edit

Service access Info
Configure the service role and EC2 instance profile that Elastic Beanstalk uses to manage your environment. Choose an EC2 key pair to securely log in to your EC2 instances.

Service role arn:aws:iam::897192298286:role/aws-elasticbeanstalk-service-role	**EC2 instance profile** aws-elasticbeanstalk-ec2-role-menna

Step 3: Set up networking, database, and tags

Edit

Networking, database, and tags Info
Configure VPC settings, and subnets for your environment's EC2 instances and load balancer. Set up an Amazon RDS database that's integrated with your environment.

Network

Step 4: Configure instance traffic and scaling

Edit

Instance traffic and scaling Info
Customize the capacity and scaling for your environment's instances. Select security groups to control instance traffic. Configure the software that runs on your environment's instances by setting platform-specific options.

Instances

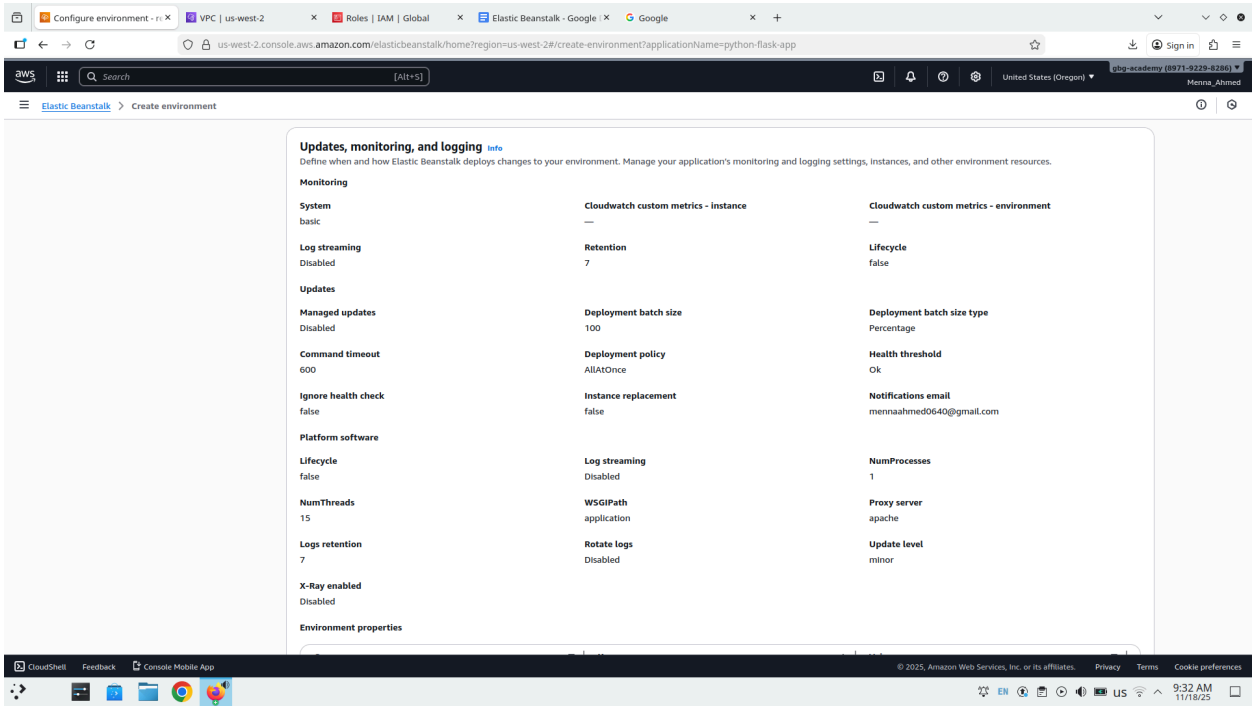
IMDSv1 Disabled	**EC2 Security Groups** sg-072f6b412781266 3f

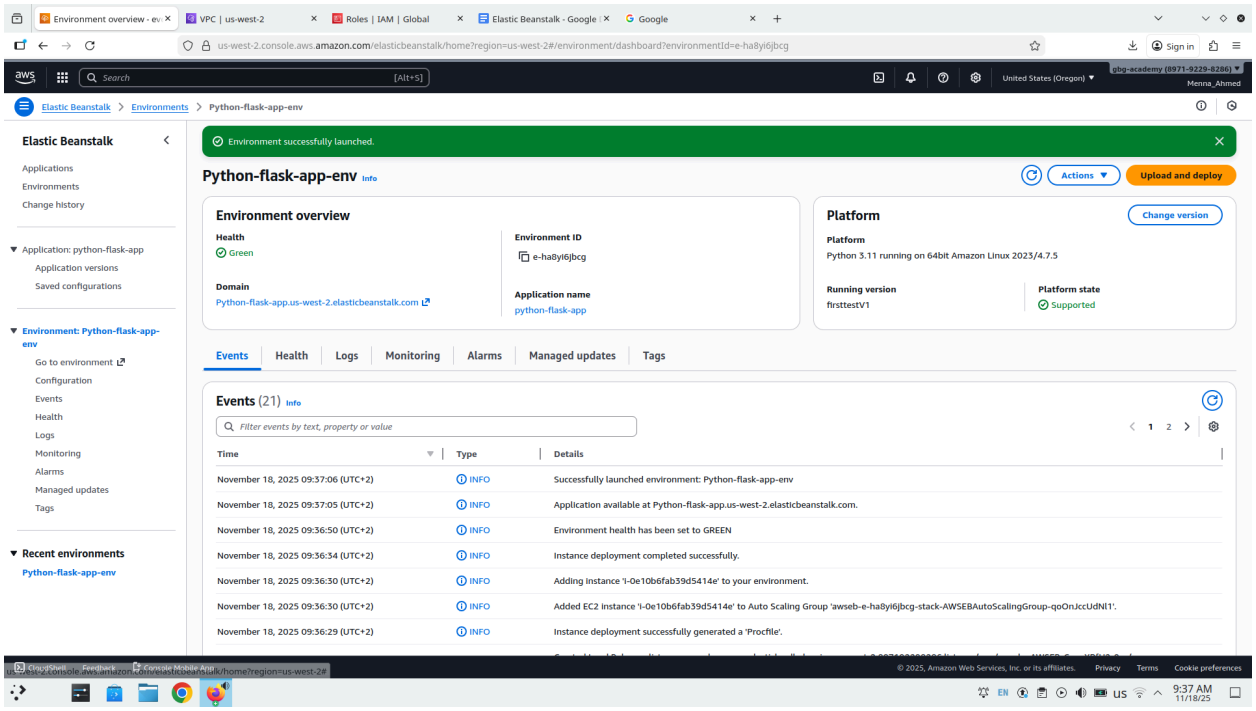
Capacity

Environment type Load balanced	**Min instances** 1	**Max instances** 2
Fleet composition On-Demand instances	**On-demand base** 0	**On-demand above base** 70
Capacity rebalancing Disabled	**Scaling cooldown** 360	**Processor type** x86_64
Instance types t3.micro	**AMI ID** ami-095ca22cd8545b80a	**Availability Zones** Any
Metric CPUUtilization	**Statistic** Average	**Unit** Bytes
Period 5	**Breach duration** 5	**Upper threshold** 6000000
Scale up increment 1	**Lower threshold** 2000000	**Scale down increment** -1

Load balancer

Load balancer visibility public	**Load balancer subnets** subnet-089884453914b91cd,subnet-02f232e411a24096f	**Load balancer type** application





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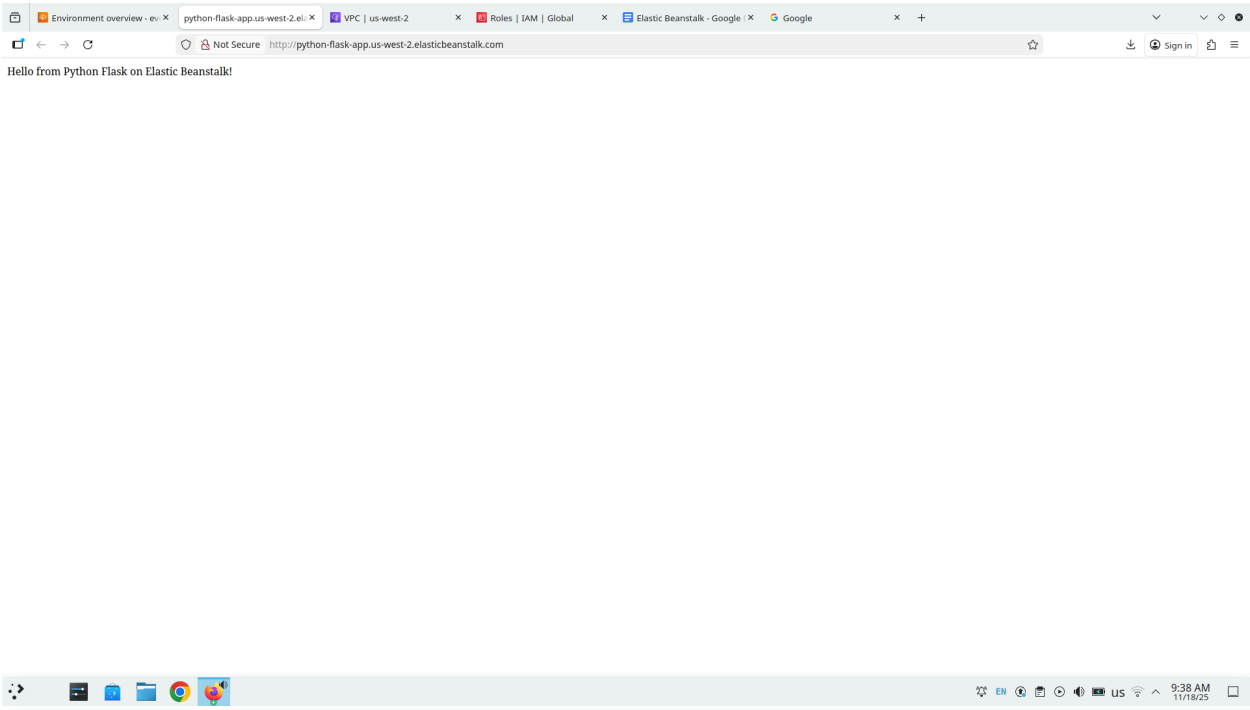

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